



UNIVERSITY OF VOCATIONAL TECHNOLOGY
Faculty of Information Technology
Bachelor of Technology in Software / Network / Multimedia and Web
Technology - 2023 / 2024 (B1, B2)
Year I – Semester - II Examination - August 2025

IT204041 – Data Structures and Algorithms

Instructions : Answer ALL Questions. Total Mark is given out of 100.

Duration : 3 hours

- (1) (a) Marina needs to automatically create usernames for a system from people's first name and the last name. She created the flowchart given in **Figure 1** to do this.

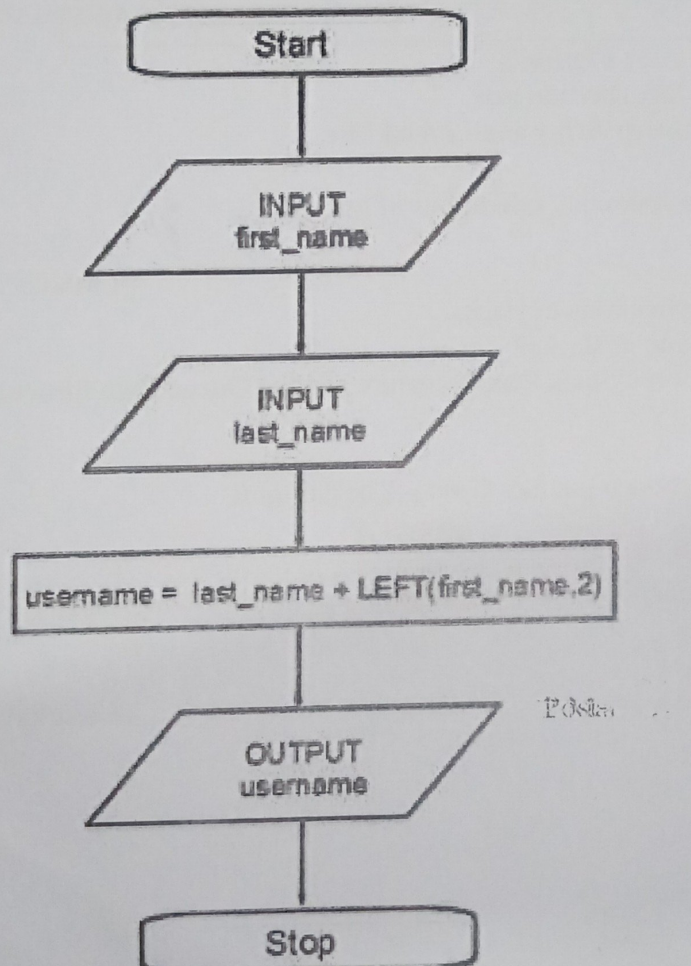


Figure 1

For example **Dan Priestley's** user name would be **PriestleyDa**. Using the process given in Figure 1, state the user name for **Graham Dodds**. (2 Marks)

- (b) Marina chooses to update the process given in part (a) to do the following.
- If the person is 18 or older, their username is their first name followed by the first

- letter of their last name.
- If the person is under 18, their username is their last name followed by the first letter of their first name. (10 Marks)
- i. Draw a flowchart for the updated process
- ii. What would be the username of a 30 year-old called Chris Watson using the updated process? (2 Marks)
- (c) Write an algorithm for a computer program that does the following
 - The user inputs three numbers as x1,x2 and x3
 - If the three numbers are the same the program outputs '**Equal**'
 - Otherwise the program outputs '**Not Equal**'. (6 Marks)

- (2) (a) What do you understand by the term **Time complexity**? (5 Marks)
- (b) Explain the importance of time complexity in terms of algorithm selection and resource allocation. (10 Marks)
- (c) State two real world applications where time complexity plays a crucial role. (5 Marks)

- (3) (a) Diagrammatically show the steps of sorting the array given in **Figure 2** using **insertion sort**.

8	3	2	7	9	1	4
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Figure 2

(10 Marks)

- (b) Write the algorithm for the insertion sort. (5 Marks)
- (c) State the differences between Array and Linked List. (5 Marks)

- (4) (a) Write algorithms for the following operations of a stack
 - i. Push()
 - ii. Pop() (4 Marks*2 = 8 Marks)
- (b) State two real-world applications of stacks. (2 Marks)
- (c) What is the LIFO principle in stacks? (5 Marks)
- (d) State the differences between Stack Data Structure and the Queue Data Structure. (5 Marks)

- (5) Write your answers by choosing either Group A or Group B

Group A

- (a) List the nodes of the Binary Tree given in **Figure 3** in the following order
 - i. In-order traversal
 - ii. Pre-order traversal
 - iii. Post-order traversal (4 Marks*3=12 Marks)

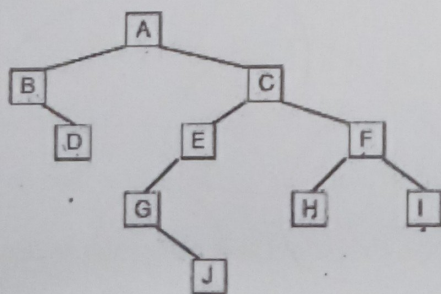


Figure 3

- (b) Write an algorithm for the Search operation of a **Binary Search tree**. (8 Marks)

- (a) Suppose you need to find the path from the root node to a leaf node that produces the largest sum of nodes (**Figure 4**).

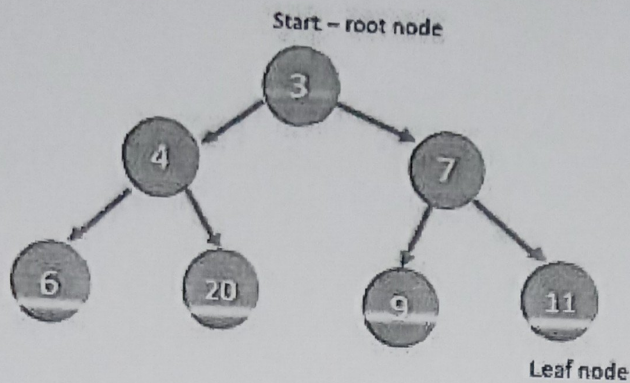


Figure 4

- Draw the Greedy Choice
 - Draw the Best Choice
- (b) Write an algorithm to **initialize a Queue**.
- (c) Write an algorithm to **check if a Queue is Full**.

(5 Marks*2=10 Marks)

(5 Marks)

(5 Marks)